Egg Segmentation and Individual Video Tracking

Computer vision internship task | Technical report

Dataset: Fire / egg-vzhtr, Roboflow version 1. Dataset source.

Executive summary

Pretrained Grounding DINO + SAM generated reference instance masks. A classical multi-threshold region-proposal / GrabCut pipeline, followed by relative-area filtering, achieved **0.5719 mean IoU and 0.6670 Dice** on 22 evaluation images. This is moderate agreement with SAM pseudo-labels, not measured true segmentation accuracy.

Deliverable	Method / verified run result
Reference segmentation	Grounding DINO localization; SAM box-prompted instance masks.
Dataset protocol	110 images; 88 development / 22 evaluation; 109 automatic groups.
Final image segmentation	Multi-threshold color/intensity regions + local GrabCut + 0.25 relative-area filter.
Final video tracking	Separate full-frame color/shape/local-contrast detector + Hungarian association; 1,440 frames at 24 FPS.
Primary limitation	Reference errors and source dependence; background confusion and missing/fragmented eggs; video identity fragmentation.

Evidence and scope

Principal experiment tables and final metrics are extracted from the supplied report_final.pdf. Earlier recorded baseline and stage diagnostics are retained for context. Figures are reused from the supplied report. No models or experiments were rerun to produce this document. Long hash dictionaries are retained in notebook/JSON outputs, not printed here.

Protocol disclosure: follow-up development occurred after the initial baseline evaluation was viewed. Candidate selection used development scores, but the follow-up evaluation is not a fresh untouched test.

Final video: Open Google Drive tracking output. Public accessibility must be checked by the applicant.

Reading guide: methods and protocol; experiment progression; quantitative and qualitative evidence; video method; reproducibility; compact full experiment appendix.

1. Reference masks and development protocol

Task 1: pretrained reference generation

Grounding DINO (**IDEA-Research/grounding-dino-tiny**) searches for *an egg.* with box threshold 0.30 and text threshold 0.25. SAM (**facebook/sam-vit-base**) receives each box and predicts three candidate masks. The highest predicted mask IoU selects the candidate. Higher-confidence masks own overlapping pixels; masks with more than 80% overlap with already assigned foreground are treated as near duplicates.

The result is an original-resolution integer mask: 0 denotes background and positive labels denote instances. SAM confidence is **not** measured segmentation IoU. Reference generation ran for all images; development examples were inspected, but a complete per-image correction record is not documented.

Task 2: approximately independent development/evaluation splits

Setting	Value / implementation
Image counts	110 total; 88 development; 22 evaluation (exact 4:1).
Seed / grouping	Seed 42; 109 groups; pHash Hamming threshold 6.
Grouping evidence	Exact duplicates, similar pHashes including horizontal/vertical flips, source stems stripped of Roboflow hash suffixes, optional manually supplied source groups.
Assignment rule	Union-find preserves connected groups. Subset-sum selects image count closest to 20% without breaking groups.
Selection rule	Highest development macro IoU; Dice breaks ties; earlier trial order breaks remaining ties.

The original train/valid/test folder names do not define the assignment split. Grouping reduces obvious duplicate leakage, but pHash can miss related capture sessions with different framing. A comprehensive manual capture-session audit is not documented.

Reference quality matters

One qualitative example reproduced later appears to include a red support in the SAM reference. This is a suspected pseudo-label error, not a confirmed corrected annotation. Reported metrics are preserved. Correcting references would require a documented audit and regenerated development/evaluation outputs.

Why this matters: agreement with an imperfect reference can penalize a correct mask or reward an incorrect one. Independent human masks and genuinely unseen source groups would provide stronger evidence.

2. Final classical image algorithm

Pipeline: image-derived region proposals → ellipse validation → local GrabCut refinement → component/duplicate validation → relative-area instance filtering.

Stage	What is computed	Why it is needed
1. Region proposals	Lab L/a/b, HSV value and inverse saturation are thresholded at quantiles 30,45,60,75,85, using both polarities.	Multiple contrasts recover shells that global whiteness or incomplete Canny contours miss.
2. Proposal validation	Opening/closing; contour area/aspect checks; fitted-ellipse mask agreement ≥0.65; overlapping proposals suppressed.	Reduce irregular and redundant regions before expensive refinement.
3. GrabCut	Ellipse interior gives foreground seeds; exterior gives background seeds; local RGB appearance is separated iteratively.	Refine boundaries from local appearance rather than keep an entire threshold component.
4. Refined instances	Keep the seed-centered connected component, fill holes, reject implausible expansion/collapse and shapes, suppress duplicates.	Prevent refinement from becoming an arbitrary background region.
5. Area filter	Retain each instance with area ≥0.25 times the largest predicted instance.	Remove numerous small false-positive regions; does not force a single egg.

Active final image parameters

Parameter	Value
Processing resolution / candidates	Maximum side 640 pixels; at most 12 proposals.
Proposal area / axis ratio	0.003–0.35 of image area; fitted axis ratio ≤2.5.
Thresholds / ellipse agreement	Five fixed quantiles; both polarities; region/ellipse IoU ≥0.65.
GrabCut	4 iterations; local ROI expansion 1.6; foreground core scale 0.35; outer probable-background ellipse scale 1.20.
Refinement validation	Area ratio 0.30–1.80; solidity ≥0.75; circularity ≥0.35.
Final area rule	Absolute area fraction 0.0; relative fraction 0.25.

The region configuration retains Canny and edge-support fields inherited from the earlier ellipse experiment. Those fields are **not used by the final region-proposal generator**; they must not be described as active edge thresholds.

SAM masks and Grounding DINO boxes are not inputs to the classical predictor. They are used only as references when measuring development/evaluation agreement. Predicted instance masks are resized to original resolution with nearest-neighbor interpolation.

3. Experiments: how the method evolved

Development stage	IoU	Dice	Count MAE	Decision
Otsu brightness baseline	0.3094	0.3721	1.0795	Simple baseline; not best.
Strict HSV	0.3489	0.4176	2.3068	Best initial baseline.
HSV without watershed	0.3487	0.4175	1.4773	Count improves; foreground problem remains.
Best contour ellipse + GrabCut	0.3449	0.3718	1.3295	Failed to beat initial IoU.
Multi-threshold regions + GrabCut	0.4752	0.5996	5.8977	Better recovery; many extra regions.
Final relative-area filter (0.25)	0.5676	0.6647	0.8864	Selected by declared dev rule.

What the diagnostics established

HSV stage failure: in three development zero-IoU examples, raw threshold recall was 0.995, 0.906 and 0.626; after cleanup it was 0.995, 0.906 and 0.769. Precision remained low because the egg joined large background regions. Final filtering removed all retained egg pixels. These diagnostic values were recorded during development; they are not new measurements from this PDF rebuild.

Watershed ablation: disabling watershed reduced count MAE by about 36%, but IoU barely changed and both variants had 12 empty predictions. Foreground extraction was the larger problem, with additional over-splitting from watershed.

Contour ellipse failure: examples showed no proposal on an egg or proposals on blue/background fragments. GrabCut cannot recover an egg that was never proposed.

Region proposal success and failure: region/GrabCut eliminated empty development predictions, but 87/88 images over-counted instances. Many extra regions were outside SAM foreground. Relative-area filtering improved precision while largely retaining recall.

Trade-off: relative_050 produced lower count MAE (0.3977), but lower IoU (0.5461) and recall (0.6246). It was not selected because macro IoU was the primary objective. The final rule can still remove genuine smaller eggs.

Appendix A–C retains all 26 principal trials with parameters/deltas, motivations, development metrics and decisions. Diagnostic ablations are documented above; JSON/CSV outputs preserve complete per-image records.

4. Follow-up evaluation: results and interpretation

Metric	Initial HSV baseline	Final classical image method
Macro foreground IoU	0.3366	0.5719
Macro Dice	0.4175	0.6670
Macro precision	0.4025	0.6670
Macro recall	0.4939	0.7118
Boundary F1 (2 pixels)	0.2050	0.4418
Micro IoU	0.3474	0.5086
Instance-count MAE	2.2727	0.7273
Empty final predictions	Not listed in source summary	0 / 22

Measured improvement: macro IoU increased by 0.2353 (23.53 percentage points). The final result remains moderate; do not state that 57.19% of eggs were correctly detected.

Uncertainty: group bootstrap with 2,000 draws gives IoU interval **0.4388–0.6976**, with 22 eval groups. Small sample size and imperfect source independence limit conclusions. Bootstrap does not remove protocol or pseudo-label bias.

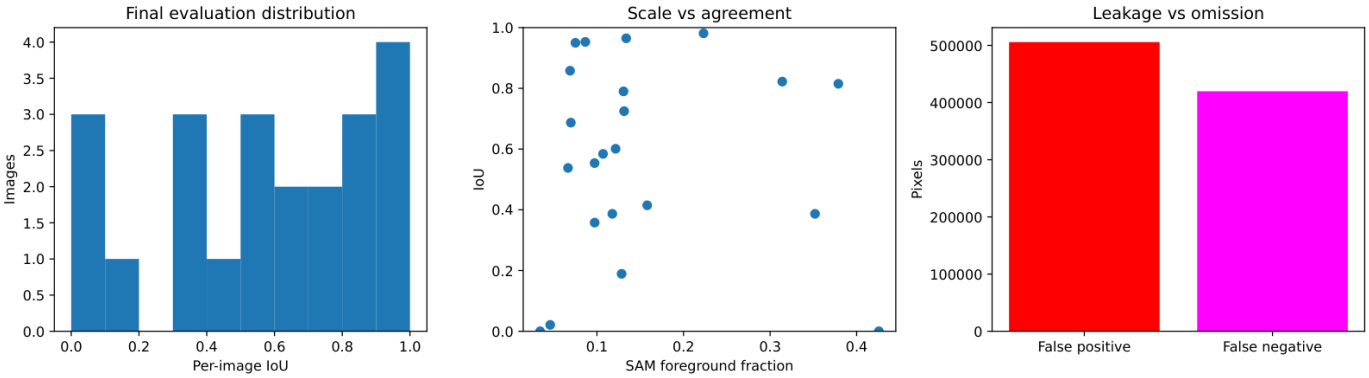


Figure 1. Original final-evaluation plots: per-image IoU distribution, foreground size versus agreement, and pooled false-positive/false-negative pixels.

Macro versus micro: macro IoU weights every image equally; micro IoU pools pixels. Lower micro IoU means aggregate pixel agreement is weaker. It is consistent with uneven performance across image/object sizes, but the figure alone does not establish a cause.

Definitions: IoU=TP/(TP+FP+FN); Dice=2TP/(2TP+FP+FN); precision=TP/(TP+FP); recall=TP/(TP+FN). Boundary F1 matches edges within 2 pixels. Count MAE is neither instance AP nor identity accuracy. Both empty masks score IoU/Dice/boundary F1=1; one empty mask scores these=0. Recall is 1 for empty truth; precision is 0 for empty prediction against nonempty truth.

5. Qualitative failures and reference uncertainty

Panels: original image | SAM reference | classical instances | error overlay. Green denotes true-positive foreground, red false positives, magenta false negatives **relative to SAM**. Filenames remain only in figure labels for sample traceability.

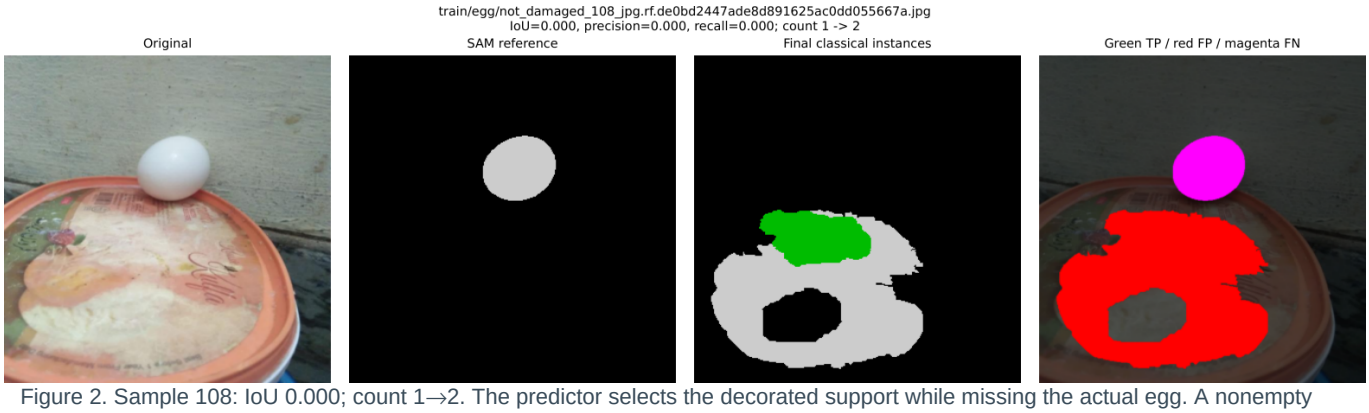


Figure 2. Sample 108: IoU 0.000; count 1→2. The predictor selects the decorated support while missing the actual egg. A nonempty output is not evidence that an egg was detected.

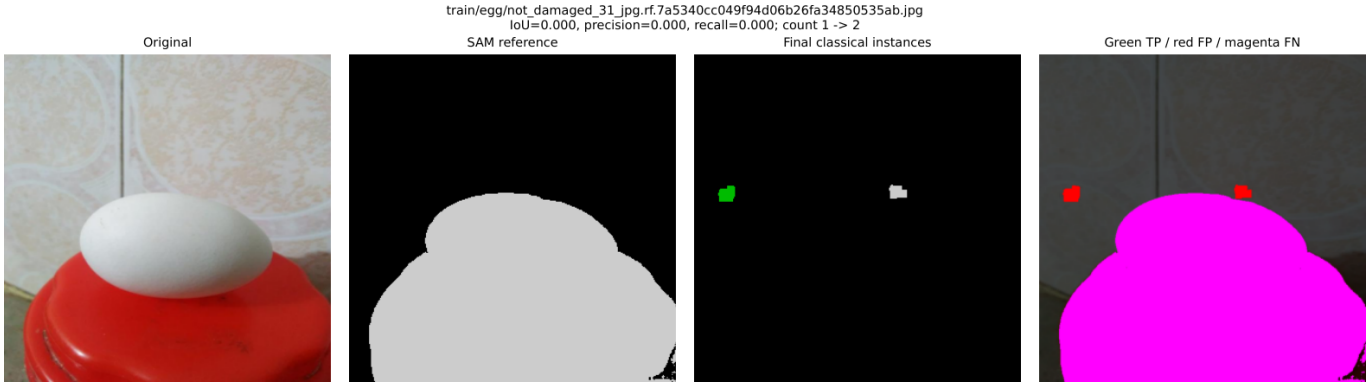


Figure 3. Sample 31: reported IoU 0.000; count 1→2. The SAM reference appears to cover both the egg and much of its red support. This is a suspected reference-mask error requiring review; the small prediction regions also fail to recover the visible shell.

Interpretation: the first example demonstrates background confusion. The second demonstrates why pseudo-ground-truth needs human auditing. Do not silently correct this figure or reinterpret the reported metric as human-annotation accuracy.

Next controlled improvement: audit references before further tuning, document every correction, regenerate scores and freeze a genuinely independent test source. Reference QA and model-development decisions must remain distinguishable.

6. Mixed-instance failures and successful examples

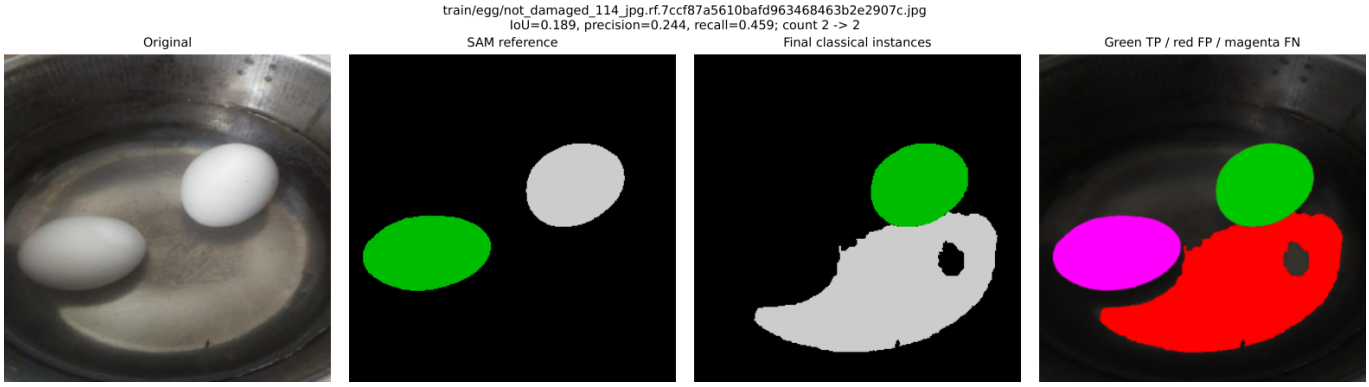


Figure 4. Sample 114: IoU 0.189, precision 0.244, recall 0.459; count 2→2. One egg is recovered, another is missed, and a bowl region is accepted. Correct counts can conceal incorrect instance correspondence.

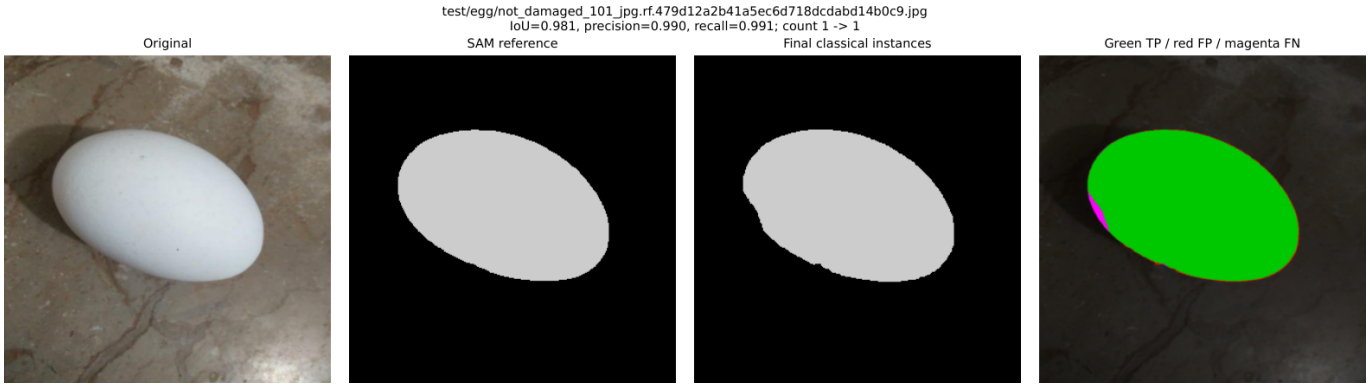


Figure 5. Sample 101: IoU 0.981, precision 0.990, recall 0.991; count 1→1. An isolated, visible egg with suitable local appearance produces close agreement with the reference.

Additional source figures show sample 42 with IoU 0.021 and count 1→5 (shell omission/background fragments), and sample 143 with IoU 0.965 and count 1→1. The original report retains those visualizations.

Overall gap: the method is much more reliable on clean, complete shells than on bright supports, cluttered trays, mixed scales or incomplete boundaries. Relative filtering improves false positives but does not solve candidate completeness.

7. Video-specific detection and individual tracking

The supplied video is 60 seconds, 640×360, 24 FPS, with 1,440 decoded frames. It includes titles, the egg demonstration, hand occlusion and explanatory screens. Framing changes made one fixed horizontal ROI unsuitable; the final detector is full-frame.

Final video stage	Active settings
Appearance	HSV value ≥165; saturation ≤95 (OpenCV 0–255 scale); 5×5 Gaussian blur; 3×3 opening/closing.
Geometry	Area fraction 0.0015–0.045; solidity ≥0.80; circularity ≥0.45; axis ratio ≤2.5; ellipse IoU ≥0.80.
Local validation	Median value contrast ≥80; interior P90–P10 brightness spread ≤65; surrounding ring has ≥80% pixels with value <150.
Association	Translation-only velocity; Hungarian cost=0.6(1−box IoU)+0.4 normalized distance; distance gate 2.0 object diagonals; area ratio 0.25–4.0.
Missing observations	Unmatched tracks retained 12 frames (0.5 seconds); predicted positions are not drawn as confirmed detections.

Why this helped

Earlier whiteness/shape checks accepted shirt patches. The final validation asks whether a smooth bright oval is surrounded by predominantly dark pixels with strong local contrast. This uses the video's white-egg/dark-table context; it is not a universal egg detector.

What was actually observed

The saved run allocated **6 track IDs**; **678 frames had no detections**. Zero-detection frames include titles and occlusions, so that count is not a miss rate. The demonstration contains two eggs, so six allocated IDs indicate identity fragmentation and/or extra tracks; the saved aggregate alone does not distinguish the causes.

Applicant visual review reported correct egg detections with no background objects observed as eggs. Identity consistency through all movements and occlusions was not independently verified. Sampled local detector inspection found missing detections during heavy hand occlusion. No measured video precision, IDF1 or MOTA is claimed.

Output video: https://drive.google.com/file/d/1pgseTFrQJit0gOINFmvm36BPF8h0tcza/view?usp=sharing

The image evaluation scores do not apply to this separately calibrated detector. Touching/hidden eggs, reflections and camera changes remain limitations. Output audio is not preserved. Improvements should distinguish observed detections from predicted track locations and use annotated intervals to quantify identity errors.

8. Reproducibility, disclosure and interview discussion

How to reproduce

Open **final.ipynb** in Kaggle with Internet and GPU enabled. Set the extracted dataset directory and original video path in Configuration. Run cells in order; inspect reference_audit.pdf and leakage_groups.csv before confirming review flags. Execute baseline/failed experiments, region/GrabCut, area filtering, frozen image evaluation, final full-frame tracking and report/export.

After the experiments, update only REPORT_NOTES and VIDEO_OUTPUT_LINK in a separate cell; rerunning Configuration resets DATA_ROOT and notes. Preserve executed notebook, requirements/exact environment, split manifest, frozen configuration, per-image evaluation, trial logs and tracks.csv. SHA256 provenance belongs in these JSON files, not in printed report pages.

Submission checks

Publish the notebook, this PDF, setup README, requirements and experiment/evaluation logs in a public GitHub repository. Upload the final tracked MP4 to Google Drive and verify access signed out. Complete reference/source audits and ID review where feasible. No public repository URL was supplied for this report; that step remains the applicant's responsibility.

AI disclosure and ownership

An AI assistant generated and revised substantial notebook code, proposed experiments, interpreted diagnostics and prepared report code. It inspected sampled original-video frames and ran local candidate tests to inform video parameters. The applicant executed Kaggle experiments, supplied results and visually inspected tracking output. The applicant should be able to explain and modify the submitted implementation; no unperformed independent review is claimed.

Questions to prepare for

Likely question	Explanation grounded in this implementation
Why is SAM not true ground truth?	Automatic masks can include supports or miss eggs; confidence does not establish correctness.
Why did global HSV fail?	It mixed egg and background; whole-component rejection removed the egg-containing region.
What does GrabCut optimize?	An iterative foreground/background appearance and spatial-smoothness graph-cut energy, initialized by image-derived ellipse seeds.
Why relative area 0.25?	It won development macro IoU among declared trials; not chosen from evaluation scores. It can discard genuine smaller eggs.
Why are dataset and video methods different?	Video-only calibration uses smooth white eggs on a dark table; heterogeneous dataset images need broader proposals.
What do six IDs mean?	Allocated IDs are not six physical eggs. Occlusion/reassociation or spurious tracks may fragment identities.
How strong is the evaluation?	22 images, imperfect pseudo-labels, incomplete source audit and reused eval exposure limit claims.

Appendix A. Initial classical experiment log

All rows are measured on 88 development images. B-F1 denotes boundary F1. These are retained as baseline/unsuccessful experiments, not the final method.

Trial	IoU	Dice	Prec.	Recall	B-F1	Count MAE
otsu	0.3094	0.3721	0.3987	0.5039	0.1746	1.0795
hsv	0.1927	0.2277	0.2311	0.3025	0.1010	1.2386
hsv_cleanup	0.2093	0.2502	0.2495	0.3457	0.1061	1.5568
hsv_watershed	0.2095	0.2503	0.2497	0.3457	0.1065	2.3977
hsv_relaxed	0.0904	0.1142	0.0971	0.1973	0.0390	2.2386
hsv_strict	0.3489	0.4176	0.4649	0.4904	0.2012	2.3068
border	0.3293	0.4154	0.6039	0.3607	0.1700	2.4318
border_relaxed	0.1905	0.2425	0.3406	0.2249	0.0809	3.3750
kmeans	0.2895	0.3433	0.3910	0.3380	0.1709	1.3750
kmeans_small	0.2865	0.3408	0.3958	0.3380	0.1626	3.3182
hsv_fine_peaks	0.2097	0.2505	0.2500	0.3457	0.1074	2.6250
hsv_coarse_peaks	0.2093	0.2502	0.2495	0.3457	0.1061	2.0568

Family base parameters

contrast=22.0; kernel=5; max_area=0.8; max_axis_ratio=3.5; max_side=768; method=hsv; min_area=0.001; min_circularity=0.25; min_solidity=0.65; peak_fraction=0.025; saturation=0.4; split_touching=True; value=0.45

Changes, motivations and decisions

otsu: kernel=1; method=otsu; split_touching=False Brightness-only baseline. Discarded under initial development IoU selection.

hsv: kernel=1; split_touching=False Reject saturated backgrounds using a pale-shell color prior. Discarded under initial development IoU selection.

hsv_cleanup: split_touching=False Opening/closing removes noise and fills small gaps. Discarded under initial development IoU selection.

hsv_watershed: No changes from family base. Distance watershed separates touching foreground objects. Discarded under initial development IoU selection.

hsv_relaxed: saturation=0.6; value=0.3 Recover dim/brown eggs; may increase skin/background leakage. Discarded under initial development IoU selection.

hsv_strict: saturation=0.25; value=0.6 Increase white-egg precision; may omit shadows/brown shells. Kept as initial baseline; later superseded.

border: method=border Use Lab novelty against clustered border-background colors. Discarded under initial development IoU selection.

border_relaxed: contrast=14; method=border Recover weaker contrast; may admit more background. Discarded under initial development IoU selection.

kmeans: method=kmeans Select foreground-enriched Lab clusters. Discarded under initial development IoU selection.

kmeans_small: method=kmeans; min_area=0.0003 Retain smaller eggs; may retain small background objects. Discarded under initial development IoU selection.

hsv_fine_peaks: peak_fraction=0.015 Separate smaller touching eggs; may over-split. Discarded under initial development IoU selection.

hsv_coarse_peaks: peak_fraction=0.05 Reduce over-splitting; may merge neighboring eggs. Discarded under initial development IoU selection.

Appendix B. Ellipse and region/GrabCut experiments

Trial	IoU	Dice	Prec.	Recall	B-F1	Count MAE
ellipse_gc_default	0.3375	0.3632	0.3979	0.3593	0.3075	1.2614
ellipse_gc_low_edges	0.3449	0.3718	0.3872	0.3764	0.2996	1.3295
ellipse_gc_strict_support	0.2460	0.2661	0.3036	0.2594	0.2374	1.3409
ellipse_gc_wider_roi	0.3382	0.3637	0.3979	0.3601	0.3056	1.2500
multithreshold_regions_grabcut	0.4752	0.5996	0.5874	0.6910	0.2820	5.8977

Contour-ellipse family base

canny_high=100; canny_low=35; edge_tolerance=3.0; grabcut_iterations=4; max_area_fraction=0.35; max_axis_ratio=2.5; max_candidates=6; max_side=640; min_area_fraction=0.003; min_edge_support=0.35; roi_expansion=1.45

ellipse_gc_default: No changes from family base. Edge-supported ellipse proposals followed by local GrabCut. Not selected among contour candidates.

ellipse_gc_low_edges: canny_high=65; canny_low=20 Recover weaker shell boundaries; may admit background contours. Best contour candidate, but dev IoU did not beat strict HSV.

ellipse_gc_strict_support: min_edge_support=0.5 Reject weak ellipse proposals; may miss partly obscured eggs. Not selected among contour candidates.

ellipse_gc_wider_roi: roi_expansion=1.8 Provide more local background samples for GrabCut. Not selected among contour candidates.

Region-proposal variant

canny_high=100; canny_low=35; edge_tolerance=3.0; grabcut_iterations=4; max_area_fraction=0.35; max_axis_ratio=2.5; max_candidates=12; max_side=640; min_area_fraction=0.003; min_edge_support=0.2; roi_expansion=1.6

Proposal quantiles: [30,45,60,75,85]; region/ellipse IoU threshold 0.65; threshold channels L/a/b/value/inverse saturation and both polarities. The inherited Canny/edge-support fields above are inactive in this variant.

Generate region proposals from several color/intensity thresholds instead of relying only on incomplete edge contours. Retained as the final proposal/refinement core, with area filtering added to reduce over-counts.

Measured runtime context

Source report lists average segmentation time of 0.4158 seconds/image for default contour ellipse + GrabCut and 2.1631 seconds/image for unfiltered multi-threshold regions + GrabCut. These are run-specific, not portable throughput guarantees. Cached-mask filter runtime was not measured.

Appendix C. Area filtering and final selection

Trial	IoU	Dice	Prec.	Recall	B-F1	Count MAE
unfiltered	0.4752	0.5996	0.5874	0.6910	0.2820	5.8977
absolute_005	0.5094	0.6254	0.6399	0.6875	0.3354	2.1477
absolute_010	0.5317	0.6413	0.6732	0.6871	0.3618	1.1705
absolute_020	0.5496	0.6516	0.7106	0.6815	0.3860	0.7500
absolute_030	0.5535	0.6474	0.7218	0.6614	0.3921	0.4773
relative_025	0.5676	0.6647	0.7207	0.6861	0.4053	0.8864
relative_050	0.5461	0.6358	0.7260	0.6246	0.3922	0.3977
combined_010_025	0.5675	0.6647	0.7207	0.6861	0.4047	0.6477
combined_020_025	0.5657	0.6623	0.7296	0.6815	0.4030	0.5795

Parameters and trade-offs

unfiltered: min_area_fraction=0.0; relative_area_fraction=0.0. Discarded under the same declared selection rule.

absolute_005: min_area_fraction=0.005; relative_area_fraction=0.0. Discarded under the same declared selection rule.

absolute_010: min_area_fraction=0.01; relative_area_fraction=0.0. Discarded under the same declared selection rule.

absolute_020: min_area_fraction=0.02; relative_area_fraction=0.0. Discarded under the same declared selection rule.

absolute_030: min_area_fraction=0.03; relative_area_fraction=0.0. Discarded under the same declared selection rule.

relative_025: min_area_fraction=0.0; relative_area_fraction=0.25. Selected by development macro IoU, then Dice and earlier trial order.

relative_050: min_area_fraction=0.0; relative_area_fraction=0.5. Discarded under the same declared selection rule.

combined_010_025: min_area_fraction=0.01; relative_area_fraction=0.25. Discarded under the same declared selection rule.

combined_020_025: min_area_fraction=0.02; relative_area_fraction=0.25. Discarded under the same declared selection rule.

Each candidate must pass both the absolute image-area fraction and relative fraction of the largest predicted instance. No SAM overlap is used for acceptance. Several qualifying instances can remain; the algorithm does not force one egg.

Reproducibility versus report readability

Complete per-image filenames, mask hashes, split assignments and tracks are retained in split.json, frozen_config.json, evaluation.json/CSV and tracks.csv. This PDF intentionally prints only technical settings and representative sample names. It is a reformatted analysis of the original measured run, not a new experimental run.